

RATHINAM TECHNICAL CAMPUS (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING / IT

Eachanari, Coimbatore — 641 021 | Affiliated to Anna University, Chennai

CAPSTONE PROJECT 70% ENGINEERING MILESTONE EVALUATION REPORT

LATE-EVENT CORRECTION & DAILY REPORTING SYSTEM: A STATEFUL WATERMARK AND DELTA RECALCULATION ENGINE FOR INSTITUTIONAL BIG DATA

Degree / Program:	B.E. Computer Science & Engineering	Evaluation Stage:	70% Milestone Review
Academic Year:	2025 – 2026	Database Engine:	SQLite 3 with Write-Ahead Logging (WAL)
GitHub Repository:	github.com/nishams63/university_data	Submission Date:	September 28, 2026

1. Abstract & Executive Summary

In institutional university administration at **Rathinam Technical Campus**, accurate operational reporting across biometric turnstile attendance, Continuous Internal Assessments (CIA), online LMS course engagement, and campus placement drives is critical for administrative governance and accreditation. However, real-world edge IoT turnstiles, departmental offline batch syncs, and delayed grade approvals produce out-of-order event arrivals. Traditional midnight snapshot architectures freeze daily reports permanently, resulting in severe temporal distortions (e.g. false absent tags). This project implements a **Stateful Watermark and Delta Recalculation Engine** that re-evaluates historical reports based on physical event timestamps, extracts incremental deltas in $O(1)$ time, bumps report versions monotonically, gates high-impact alterations ($\geq 15\%$ drift or grade flips) into a human review queue, and maintains an immutable audit ledger with 1-click compensating rollbacks. At this 70% milestone, the system has achieved 100% automated test pass (23/23 tests), verified 100% exact match convergence against ground truth (MAE = 0.000), and proved sub-40ms latencies with peak RAM usage strictly under 1.0 MB.

2. Resolution of Prior Evaluator Criticisms (35% → 70%)

Evaluator Criticism	Root Cause in 35% Baseline	Engineering Resolution (70% Prototype)	Status
1. Schema & Concurrency Controls	Default SQLite rollback journal; no foreign keys; unconstrained version increments.	Enforced PRAGMAs: journal_mode=WAL, busy_timeout=10000ms, foreign_keys=ON. Added composite unique constraints (reporting_date, domain) and (report_id, version_number).	RESOLVED (5/5 Tests)
2. Performance Evaluation	Theoretical claims; lack of execution-timed latency distributions and memory profiling.	Built benchmark engine with time.perf_counter() and tracemalloc. Benchmarked 100 to 5000 events and 4 replay workloads (4000 events total).	RESOLVED (Empirical)
3. Transactional Boundaries	Ingestion, recalculation, and versioning ran in disjoint, uncoordinated queries.	Built ingest_and_process_atomic() wrapping ingestion, delta calculation, versioning, and audit logging into single atomic transaction boundary with backoff retry.	RESOLVED
4. Recalculation Complexity	Executed $O(N)$ table scans and JSON deserialization per incoming late event.	Engineered $O(1)$ stateful delta contribution extraction (delta = f(domain, payload)), reducing latency to 21-38ms even under 5,000 events.	RESOLVED
5. Version Lineage & Review UX	No version lineage visualization; reviews lacked before/after comparison diffs.	Added Version Lineage Timeline (v1->v2->v3), Before/After diff visualizer cards, explanation panels, confirmation modals, and Engine Diagnostics tab.	RESOLVED

3. Empirical Performance Benchmark Results (Hardware-Timed)

Benchmarks were executed using Python's high-resolution timer (time.perf_counter()) and heap allocation profiler (tracemalloc) against an isolated SQLite WAL database instance.

A. Event Scaling Benchmarks (100 to 5,000 Events)

Event Load	Duration	Throughput	Avg Latency	Median (P50)	P95 Latency	P99 Latency	Peak RAM	Exact Match
100 Events	2.30 s	43.51 eps	22.98 ms	21.80 ms	36.88 ms	137.14 ms	0.98 MB	100.0%
500 Events	10.58 s	47.28 eps	21.14 ms	20.50 ms	33.66 ms	46.44 ms	0.44 MB	100.0%
1,000 Events	21.79 s	45.90 eps	21.78 ms	20.64 ms	37.26 ms	52.97 ms	0.45 MB	100.0%
5,000 Events	192.85 s	25.93 eps	38.56 ms	32.77 ms	70.80 ms	116.05 ms	0.57 MB	100.0%

B. Historical Replay Benchmarks (4 Distinct Workload Profiles, 4,000 Events)

Workload Profile	Characteristics	Events	Duration	Throughput	Avg Latency	P95 Latency	Peak RAM
1. Normal Stream	10% late, 3% dup, 1% inv	1,000	31.53 s	31.72 eps	31.52 ms	53.12 ms	0.419 MB
2. Late-Heavy	40% late arrivals (sync lag)	1,000	7.50 s	133.33 eps	7.49 ms	11.73 ms	0.065 MB
3. Duplicate-Heavy	30% duplicate retries (storm)	1,000	6.11 s	163.63 eps	6.10 ms	7.96 ms	0.064 MB
4. 7-Day Delayed Lag	35% late, 72h-168h delays	1,000	6.51 s	153.66 eps	6.50 ms	9.22 ms	0.065 MB
Cumulative Total	4 Workload Profiles Replayed	4,000	51.65 s	77.45 eps	12.91 ms	20.51 ms	0.419 MB

4. Mathematical Reconciliation & Error Elimination Across 6 Scenarios

Evaluation Scenario	Late Ratio	Baseline MAE	Corrected MAE	Baseline RMSE	Corrected RMSE	Exact Match %
Scenario 1: 0% Late Events (Ideal)	0.0%	53.37	0.000	93.50	0.000	100.0%
Scenario 2: 5% Late Events (Mild)	5.0%	55.36	0.000	101.67	0.000	100.0%
Scenario 3: 10% Late Events (Standard)	10.0%	58.42	0.000	119.67	0.000	100.0%
Scenario 4: 25% Late Events (Outage)	25.0%	100.04	0.000	213.72	0.000	100.0%
Scenario 5: Duplicate-Heavy (30% Dup)	15.0%	70.48	0.000	130.46	0.000	100.0%
Scenario 6: Very Late (Multi-Day Lag)	30.0%	117.25	0.000	222.37	0.000	100.0%

5. Automated Test Suite & Quality Assurance (23/23 Passing)

The codebase maintains 100% automated test pass rate across 23 test functions organized into 8 focused modules:

Test Module	Tests	Key Verification Scope
test_concurrency.py	5	Thread-safe duplicate races, concurrent unique events, concurrent late events targeting identical historical dates, and review approval state locks.
test_database_hardening.py	5	Verifies SQLite WAL mode, foreign key enforcement, composite unique constraints (reporting_date, domain) and (report_id, version_number), and 24h lateness boundaries.
test_failure_injection.py	2	Injects synthetic runtime exceptions during processing to verify transactional rollback and database consistency.
test_pipeline.py	4	Tests on-time ingestion, late event tagging, duplicate event ID deduplication, and invalid payload rejection.
test_correction.py	3	Tests baseline aggregation, O(1) stateful delta calculation, and high-impact (≥ 15%) review threshold routing.
test_audit_rollback.py	2	Verifies write-only audit trail logging and compensating rollback idempotency.
test_reconciliation.py	1	Proves the Ground Truth Convergence Invariant across all reporting dates.
test_integration.py	1	End-to-end user flow: ingestion -> lateness evaluation -> review approval -> report update.
Total Suite Execution	23 / 23 Passing	Execution Time: 8.90s Exit Code: 0 (100% Passing)

6. Remaining 30% Implementation Roadmap (Towards 100% Viva)

Phase / Month	Target Enhancement	Technical Scope & Deliverables
Phase 8 (Oct 2026)	Distributed PostgreSQL Adapter	Dual-driver configuration enabling zero-downtime migration from SQLite WAL to PostgreSQL + TimescaleDB for multi-campus scaling.
Phase 9 (Nov 2026)	Streaming Ingestion Gateway	Decouple HTTP gateway from database writes using Apache Kafka or Redis Streams with distributed consumers.
Phase 10 (Dec 2026)	Institutional RBAC & SSO	OAuth2 / JWT integration with role hierarchy (Dean, HoD, Faculty Coordinator, Auditor).
Phase 11 (Jan 2027)	Automated Anomaly Detection	Isolation forest ML models detecting bulk turnstile log manipulations or grade tampering bursts.
Phase 12 (Feb 2027)	Final Dissertation & Defense	Full academic thesis, viva presentation slides, and institutional deployment handbook.

7. Academic Review Committee Sign-off

This evaluation report certifies that the Capstone Project entitled "Late-Event Correction & Daily Reporting System: A Stateful Watermark and Delta Recalculation Engine for Institutional Big Data" satisfies all requirements for the 70% Engineering Milestone Review.

Project Candidate
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